HK IMO Training 2023-2024
Problem Set 18 Suggested Solutions

Problem 69 (Bosnia and Herzegovina TST 2012 Problem 4)**.** Let p be a prime number. Define a sequence $\{a_n\}$ by $a_1 = p + 1$ and

$$a_{n+1} = a_1 a_2 \cdots a_n + p$$

for $n \geqslant 1$. Find all p such that the sequence contains at least one perfect square.

Solution. Answer: $p = 3$.

If $p = 2$, we have $a_1 = 3$, which is not a prefect square. Also, we have $a_{n+1} = 3a_2 \cdots a_n + 2 \equiv 2 \pmod 3$ for $n \geqslant 1$, which cannot be a perfect square.

If $p \equiv 3 \pmod 4$, we have $a_1 \equiv 0 \pmod 4$. Thus, $a_{n+1} \equiv 3 \pmod 4$ for $n \geqslant 1$, which cannot be a perfect square. The only possible perfect square is $a_1 = p+1$. If $a_1 = m^2$, then $p = m^2 - 1 = (m-1)(m+1)$. Since p is a prime, we must have $m - 1 = 1$. This implies $m = 2$ and $p = 3$. When $p = 3$, we check that $a_1 = 4 = 2^2$.

If $p \equiv 1 \pmod 4$, we have $a_1 \equiv 2 \pmod 4$ and $a_2 = a_1 + p \equiv 3 \pmod 4$. Assuming $a_k \equiv 3 \pmod 4$ for all $k = 2, 3, \ldots, n$, we have $a_{n+1} \equiv 2 + p \equiv 3 \pmod 4$. Therefore, $a_n \equiv 3 \pmod 4$ for all $n \geqslant 2$ by induction. Thus, all a_j's (including a_1) are not perfect squares. $\square$

Problem 70 (Bosnia and Herzegovina TST 2018 Problem 4)**.** In a 1000×1000 board, each cell is coloured black or white. It is known that there exists a 10×10 sub-board such that every cell is black, and there exists a 10×10 sub-board such that every cell is white. For each 10×10 sub-board, we define its *score* to be the difference of the number of black cells and the number of white cells in this sub-board. Let N be the minimum score among all 10×10 sub-boards. Find the maximum possible value of N.

Solution. Answer: 10.

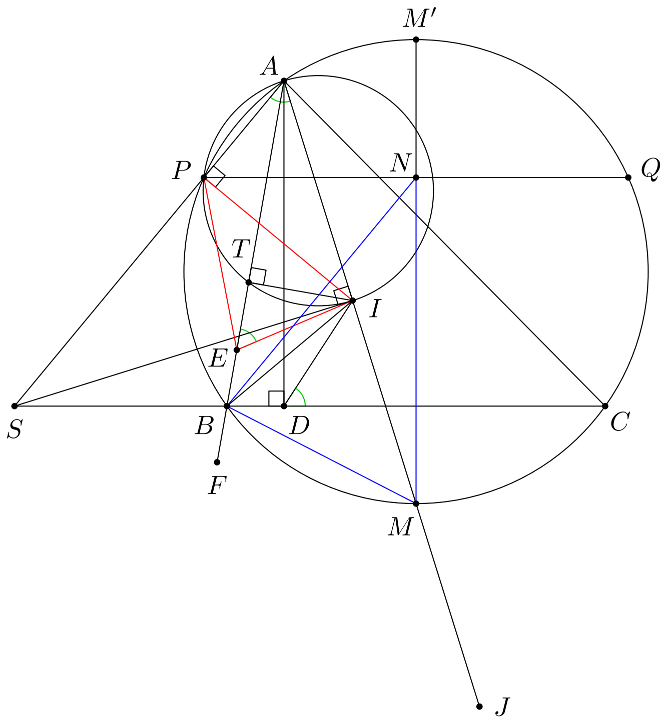
We first show that $N \leqslant 10$. Consider a 10×10 sub-board whose every cell is black. We move this sub-board step by step to the 10×10 sub-board whose every cell is white, such that it is moved by one unit horizontally or vertically in each step. Let S be the number of black cells minus the number of white cells in this sub-board (which varies at different steps). Initially, we have $S = 100$. In the end, we have $S = -100$. In each step, 10 cells of the sub-board are changed, and hence S is changed by at most ± 20. As S changes from 100 to -100, there must be a step for which $-10 \leqslant S \leqslant 10$. This implies $N \leqslant |S| \leqslant 10$.

Then we show that $N \geqslant 10$ by providing a construction. We colour all cells in the first 10 columns black. Then we colour the cells in the 11th column black and white alternately. Finally we colour all cells in the last 989 columns white. Clearly, the conditions are satisfied. Consider a 10×10 sub-board whose leftmost column is column k.

- If $k = 1$, the score is $100 - 0 = 100$.
- If $2 \leqslant k \leqslant 11$, there are $115 - 10k$ black cells and $-15 + 10k$ white cells, and so the score is $|130 - 20k| \geqslant 10$.
- If $k \geqslant 12$, the score is $100 - 0 = 100$.

Therefore, $N = 10$ in this example. $\square$

Problem 71 (China TST 2019 Test 3 Problem 5). In $\triangle ABC$, let D be the foot of altitude from A. Let E and F be two points on the line AB such that $BD = BE = BF$, where E lies between A and B. Let I and J be the incentre and A-excentre of $\triangle ABC$ respectively. Prove that there exist two points P and Q on the circumcircle of $\triangle ABC$ such that $PB = QC$ and $\triangle PEI \sim \triangle QFJ$.



Solution. Let P be the A-Sharky-Devil point of $\triangle ABC$ (the point on (ABC) such that $\angle API = 90°$), and let Q be the point on (ABC) such that $PQ \,/\!/\, BC$. Clearly, we have $PB = QC$ as $PBCQ$ is an isosceles trapezoid. We claim that $\triangle PEI \sim \triangle QFJ$.

Let M be the midpoint of arc BC of (ABC) not containing A, N be the midpoint of PQ. We first prove that $\triangle PEI \sim \triangle NBM$.

Let T be the projection of I onto AB. Notice the following facts.

- The line AP, the line BC and the line through I perpendicular to AI are concurrent at some point S, since this is the radical centre of (ABC), (API) and (BCI).
- The points A, S, D, I are concyclic since $\angle AIS = \angle ADS = 90°$.
- $\triangle BDI \cong \triangle BEI$ since BI bisects $\angle CBA$ and $BD = BE$.

It follows that

$$\angle TIE = 90° - \angle IET = 90° - (180° - \angle BEI) = 90° - (180° - \angle BDI) = 90° - \angle CDI = 90° - \angle SAI = \angle AIP.$$

Therefore, $\angle PIE = \angle AIT = 90° - \dfrac{A}{2} = 90° - \angle MBC = \angle NMB$.

Next, let M' be the midpoint of arc BC of (ABC) containing A. Then

$$NM = PM\cos\angle NMP = PM\cos\angle M'MP = PM\sin\angle PM'M = PM\sin\angle PAM = PM\sin\angle CDI.$$

Also, we have

$$IE = ID = BI\cdot\frac{\sin\frac{B}{2}}{\sin\angle IDB} = \frac{BI\sin\frac{B}{2}}{\sin\angle CDI} = \frac{r}{\sin\angle CDI}$$

where r is the inradius of $\triangle ABC$. Note that $\triangle PIT \sim \triangle PMB$ by a property of the Sharky-Devil point. This implies

$$\frac{PI}{PM} = \frac{IT}{MB} = \frac{r}{MB}.$$

Combining all these, we obtain

$$\frac{PI}{IE} = \frac{PM\cdot\dfrac{r}{MB}}{\dfrac{r}{\sin\angle CDI}} = \frac{PM\sin\angle CDI}{MB} = \frac{NM}{MB}.$$

Therefore, $\triangle PEI \sim \triangle NBM$.

Lastly, as N is the midpoint of PQ, B is the midpoint of EF, M is the midpoint of IJ, we have $\triangle PEI \sim \triangle QFJ$ (for a more detailed proof, one may use spiral similarity, for example). □

Problem 72 (China TST 2019 Test 2 Problem 3)**.** Let $n \geqslant 2$ be an even positive integer, and let $x_1, x_2, \ldots, x_n$ be nonnegative real numbers with sum 1. Find the maximum possible value of

$$\sum_{1 \leqslant i<j \leqslant n} \min \left\{(i-j)^2,(n+i-j)^2\right\} x_i x_j .$$

Solution. Answer: $\dfrac{n^2}{16}$.

When $x_1=x_{\frac{n}{2}+1}=\dfrac{1}{2}$ and $x_i=0$ for all $i \neq 1, \dfrac{n}{2}+1$, we have

$$S=\sum_{1 \leqslant i<j \leqslant n} \min \left\{(i-j)^2,(n+i-j)^2\right\} x_i x_j=\min \left\{\left(\frac{n}{2}\right)^2,\left(\frac{n}{2}\right)^2\right\} \cdot\left(\frac{1}{2}\right)^2=\frac{n^2}{16} .$$

It remains to prove that $S \leqslant \dfrac{n^2}{16}$.

Note that

$$S \leqslant \frac{n}{2} \sum_{1 \leqslant i<j \leqslant n} \min \{j-i, n-(j-i)\} x_i x_j$$

since one of $j-i$ and $n-(j-i)$ must be at most $\dfrac{n}{2}$. Let $y_i=x_i+x_{i+1}+\cdots+x_{i+\frac{n}{2}-1}$ where all indices are taken modulo n. Then

$$\sum_{1 \leqslant i<j \leqslant n} \min \{j-i, n-(j-i)\} x_i x_j=\sum_{i=1}^{\frac{n}{2}} y_i y_{i+\frac{n}{2}}$$

by expanding the right-hand side. It follows that

$$S \leqslant \frac{n}{2} \sum_{i=1}^{\frac{n}{2}} y_i y_{i+\frac{n}{2}} \leqslant \frac{n}{2} \sum_{i=1}^{\frac{n}{2}}\left(\frac{y_i+y_{i+\frac{n}{2}}}{2}\right)^2=\frac{n}{2} \sum_{i=1}^{\frac{n}{2}}\left(\frac{1}{2}\right)^2=\frac{n^2}{16} .$$

□